



Predicting stochastic-system statistics

- What are the statistics of the state of a system driven by a random process?
- That is, how do the system dynamics influence the time-propagation of $x(t)$?
 ⇒ We do not know $x(t)$ exactly, but we can develop a pdf for $x(t)$.
- First, we contrast a linear system driven by white noise $w(t)$, possibly in addition to deterministic input $u(t)$:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t), \quad x(0) \sim \mathcal{N}(\bar{x}_0, \Sigma_{\bar{x}_0}), \quad w(t) \sim \mathcal{N}(\bar{w}, \Sigma_{\bar{w}}),$$

with a deterministic model, which can be easily simulated:

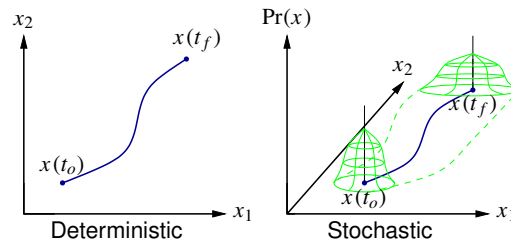
$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0, \quad x(0), u(t) \text{ known.}$$

- In the first case, the state is never known with certainty; in the second case, the system response is completely specified. Therefore, there is no uncertainty.



Visualizing the information

- In the stochastic case, the best we can do is to say how uncertainty in the state changes with time.
- Deterministic: $x(t_0)$ and inputs known
 ⇒ State trajectory is deterministic.
- Stochastic: pdfs for $x(t_0)$ and for inputs known ⇒ Can find only pdf for $x(t)$.
- We will work with Gaussian noises, which are uniquely defined by their first and second-central moments.
- The Gaussian assumption is not essential, but in any case we will only ever track the first two moments.



Discrete-time systems

- NOTATION: Until now, we have always used capital letters for RVs. The state of a system driven by a random process is an RV, so we could now call it $X(t)$ or X_k . But, it is more common to retain standard notation $x(t)$ or x_k and understand from context that we are now discussing an RV.
- We seek to find the statistics of x_k for a discrete system; we will later look at how to get an equivalent answer for continuous systems.
- Model (where A_d and B_d are assumed known and can be time-varying):

$$x_k = A_{d,k-1}x_{k-1} + B_{d,k-1}u_{k-1} + w_{k-1},$$

where

x_k : State vector, a random process.

u_k : Deterministic control inputs.

w_k : Noise that drives the process (process noise), a random process.



Discrete-time systems

- We make some key assumptions about the driving noise:
 1. Zero mean: $\mathbb{E}[w_k] = 0 \quad \forall \quad k$.
 2. White: $\mathbb{E}[w_{k_1} w_{k_2}^T] = \Sigma_{\tilde{w}} \Delta(k_1 - k_2)$.
 - $\Sigma_{\tilde{w}}$ is called the spectral density of the noise signal w_k .
- We also make assumptions regarding state uncertainty at startup. Statistics of initial condition:

$$\begin{aligned}\mathbb{E}[x_0] &= \bar{x}_0; \\ \mathbb{E}[(x_0 - \bar{x}_0)w_k^T] &= 0 \quad \forall \quad k \\ \Sigma_{\tilde{x},0} &= \mathbb{E}[(x_0 - \bar{x}_0)(x_0 - \bar{x}_0)^T].\end{aligned}$$

- Key question: How do we propagate the statistics of this system?



Propagating the mean

- The mean value is:

$$\begin{aligned}\mathbb{E}[x_k] &= \bar{x}_k = \mathbb{E}[A_d x_{k-1} + B_d u_{k-1} + w_{k-1}] \\ &= A_d \bar{x}_{k-1} + B_d u_{k-1}.\end{aligned}$$

Therefore, the mean propagation is:

$$\begin{aligned}\bar{x}_0 &: \text{Given} \\ \bar{x}_k &= A_d \bar{x}_{k-1} + B_d u_{k-1}.\end{aligned}$$

- Deterministic simulation and the mean values of a stochastic simulation are treated the same way.



Propagating variations about the mean

- To study the random variations about the mean, we need to form the second central moment of the statistics:

$$\Sigma_{\tilde{x},k} = \mathbb{E}[(x_k - \bar{x}_k)(x_k - \bar{x}_k)^T].$$

- Easiest to study if we note that:

$$\begin{aligned}x_k - \bar{x}_k &= A_d x_{k-1} + B_d u_{k-1} + w_{k-1} - A_d \bar{x}_{k-1} - B_d u_{k-1} \\ &= A_d (x_{k-1} - \bar{x}_{k-1}) + w_{k-1}.\end{aligned}$$

- Thus, $\Sigma_{\tilde{x},k} = \mathbb{E}[(A_d (x_{k-1} - \bar{x}_{k-1}) + w_{k-1})(A_d (x_{k-1} - \bar{x}_{k-1}) + w_{k-1})^T]$.

- Three terms:

1. $\mathbb{E}[A_d (x_{k-1} - \bar{x}_{k-1})(x_{k-1} - \bar{x}_{k-1})^T A_d^T] = A_d \Sigma_{\tilde{x},k-1} A_d^T$.
2. $\mathbb{E}[w_{k-1} w_{k-1}^T] = \Sigma_{\tilde{w}}$.
3. $\mathbb{E}[A_d (x_{k-1} - \bar{x}_{k-1}) w_{k-1}^T] = ?$



Propagating variations about the mean

- The third term $\mathbb{E}[A_d(x_{k-1} - \bar{x}_{k-1})w_{k-1}^T]$ is a cross-correlation.
 - But, x_{k-1} depends only on x_0 and w_m for $m = 0 \dots k-2$.
 - w_{k-1} is white noise uncorrelated with x_0 .
 - Therefore, the third term is zero.

Therefore, the covariance propagation is:

$\Sigma_{\tilde{x},0}$: Given

$$\Sigma_{\tilde{x},k} = A_d \Sigma_{\tilde{x},k-1} A_d^T + \Sigma_{\tilde{w}}.$$

- In this equation, $A_d \Sigma_{\tilde{x},k-1} A_d^T$ is the homogeneous part; if the system is stable, $A_d \Sigma_{\tilde{x},k-1} A_d^T \leq 0$ (is contractive, reduces uncertainty).
- $\Sigma_{\tilde{w}}$ is the driving term, which always increases uncertainty since $\Sigma_{\tilde{w}} > 0$.



Steady-state solution

- If A_d and $\Sigma_{\tilde{w}}$ are constant and A_d is stable, there is a steady-state solution to $\Sigma_{\tilde{x},k} = A_d \Sigma_{\tilde{x},k-1} A_d^T + \Sigma_{\tilde{w}}$.
 - As $k \rightarrow \infty$, $\Sigma_{\tilde{x},k} = \Sigma_{\tilde{x},k+1} = \Sigma_{\tilde{x}}$.
 - Then, $\Sigma_{\tilde{x}} = A_d \Sigma_{\tilde{x}} A_d^T + \Sigma_{\tilde{w}}$ (a discrete Lyapunov equation. In Octave, `dlyap.m`)
- Can solve steady-state response by hand in the scalar case.
- For example, consider: $x_k = \alpha x_{k-1} + w_{k-1}$. $\mathbb{E}[w_{k-1}] = 0$; $\Sigma_{\tilde{w}} = 1$; $\bar{x}_0 = 0$.
- Then, we solve for the steady-state covariance:

$$\Sigma_{\tilde{x}} = \alpha \Sigma_{\tilde{x}} \alpha^T + 1$$

$$\Sigma_{\tilde{x}}(1 - \alpha^2) = 1$$

$$\Sigma_{\tilde{x}} = \frac{1}{1 - \alpha^2}.$$

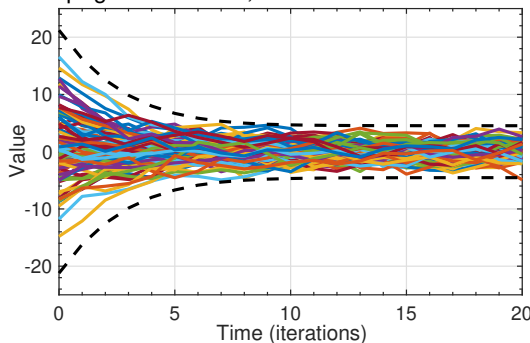
- Valid for $|\alpha| < 1$; otherwise unstable.



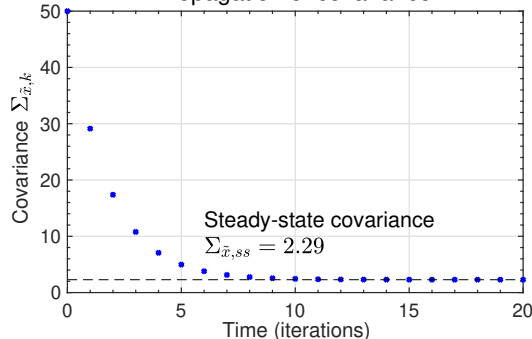
Illustrating the example: $\alpha = 0.75$ and $\Sigma_{\tilde{x},0} = 50$

- Example plots 100 random trajectories and compares propagation of $\Sigma_{\tilde{x},k}$ with steady-state `dlyap.m` solution.

Propagation of state, with confidence bounds



Propagation of covariance



- The error bounds plotted are 3σ bounds (i.e., $\pm 3\sqrt{\Sigma_{\tilde{x},k}}$).



Summary

- With deterministic state-space systems, we can simulate a model and have no uncertainty regarding the state trajectory.
- With stochastic state-space systems, we instead track the mean and covariance of the RV representing the state at every point in time.
- You learned how to update the mean and covariance:

$$\begin{aligned}\bar{x}_k &= A_d \bar{x}_{k-1} + B_d u_{k-1}. \\ \Sigma_{\tilde{x},k} &= A_d \Sigma_{\tilde{x},k-1} A_d^T + \Sigma_{\tilde{w}}.\end{aligned}$$

- The true state x_k is expected to be within $\bar{x}_k \pm 3\sqrt{\text{diag}(\Sigma_{\tilde{x},k})}$ approximately 99.7 % of the time if the model is correct and noises are Gaussian.